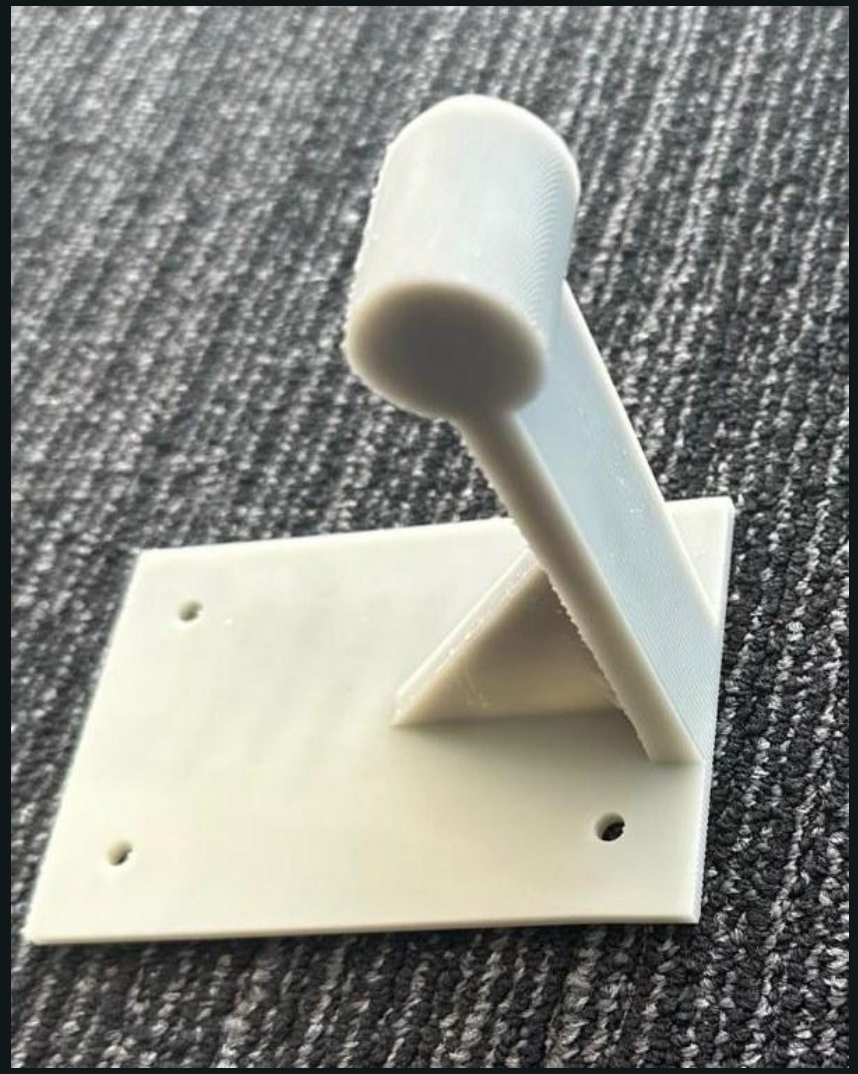
CoForm

AI specialists for custom 3D-printed parts

A simple holder needs decisions about fit, strength and printing.

We built CoForm so a person can describe a part, inspect a checked design, and prepare it for fabrication.



Designed by our AI agents, printed in resin

Manufacturing & Materials MIT Media Lab November 1, 2026

One request becomes a checked design

"Build a water bottle holder which can attach to a desk and hold it."

1 Describe the need

The user gives a request and any known sizes or load limits.

2 Choose tested components

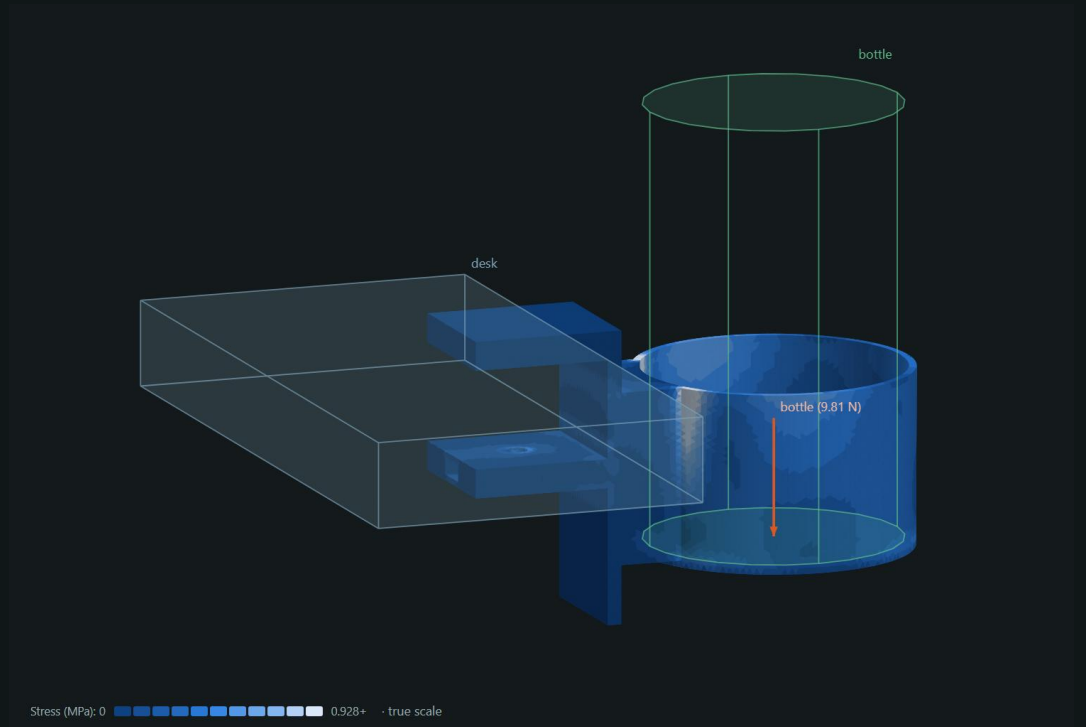
Mechanics chooses the clamp and cup. Fabrication and measurement review the concept.

3 Build, review and refine

Code assembles and tests the geometry. Agents review failures and propose improvements.

4 Inspect, print and measure

The user authorizes the print, starts the printer, then enters the measured bend.



Tested assumptions: 25 mm desk, 75 mm bottle diameter, 1 kg load. The request supplied no sizes.

The three agents and where they come from

CoForm creates three separate AI roles. In the demonstrated runs, all three use NVIDIA Nemotron 3 Ultra through NVIDIA's API.

Agent	Responsibility	Contribution to the design
Mechanics	Will it hold the load?	Chooses the mechanism and proposes changes to strength and stiffness.
Fabrication	Can we print it?	Reviews resin drainage, supports and clearances. Proposes material savings.
Measurement	Can we test it?	Plans how to hold, load and measure the part. Does not change geometry.

Each role gets its own instructions and model call. They coordinate through a shared record of proposals, reviews and measured results.

Why a collective: the requirements compete

Saving resin can make a part bend too far. Separate roles keep strength, fabrication and testability visible during the same decision.

A real decision on the backpack holder

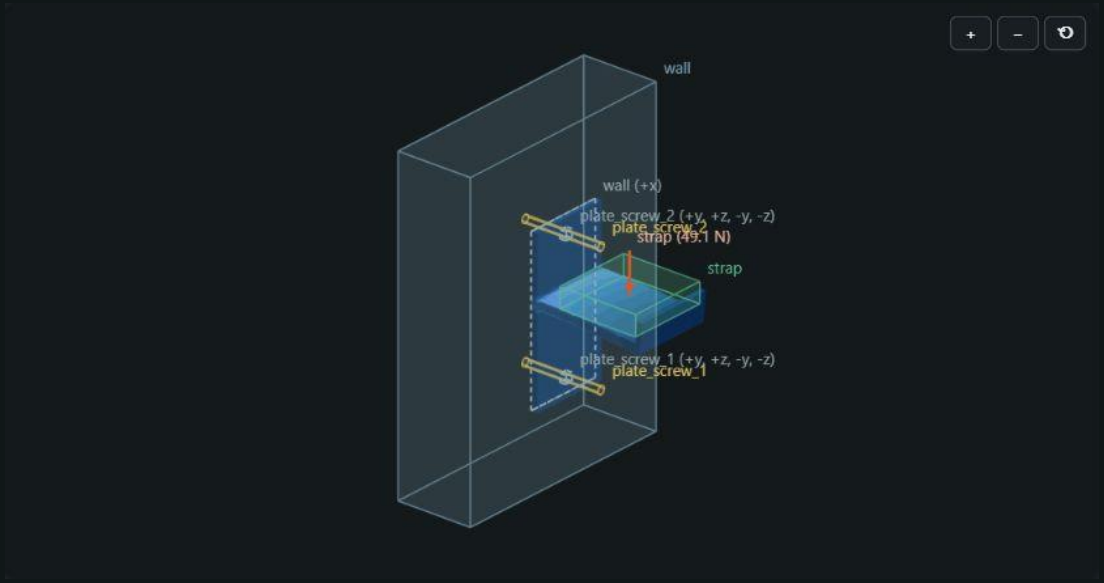
Proposal	Predicted bend	Decision
Mechanics: lighter geometry	2.52 mm	Reject: exceeds 2 mm limit
Fabrication: moderate thinning	1.81 mm	Adopt: 99.1 g and passes

Agents propose and review. Code tests the alternatives and selects the lightest passing design. A verified failure cannot be voted away.

There is no lead AI. Fixed software stages coordinate the roles. This example shows a useful contribution, not proof that three agents beat one.

Latest hook: 48% less modeled material

Request: a wall hook for a backpack. The current component workflow produced this design.



This newer hook passes the modeled checks. Print preparation is still pending.

5 kg modeled load. Bend limit: 2 mm. Stress limit: 23 MPa.

Prediction	First accepted	Best design
Part mass	44.2 g	22.9 g (-48%)
Bend	0.21 mm	1.87 mm
Peak stress	3.6 MPa	15.7 MPa

An earlier 99.1 g backpack holder has finished printing. Its physical load test is still pending.

The evidence supports a prototype, with limits

A working request-to-print workflow

It creates geometry, checks fit, runs physics simulations and prepares prints in Formlabs PreForm.

348 automated tests cover the software workflow. One design has finished printing.

Physical load testing and measurement-based calibration are still pending.

A collective advantage is still unproven

Beam-sizing benchmark: 4 runs per method. All passed the simulated acceptance checks.

Method	Material vs reference
Numerical optimizer	76%
Single AI agent	88%
Three-agent collective	93%

Lower is better. The optimizer won this narrow task. Full protocol and costs: slide 9.

CoForm makes the design decision inspectable

A request produces a design, the reasons behind it, and the evidence needed to decide what to print and test.

Next physical step

Load the printed backpack holder, measure its bend, and recheck the design with corrected material stiffness.

Next research step

Compare one agent and the collective on the same open-ended requests, tools, checks and budget.

Code and evidence: ScienceClaw_Project / Jash_ver

What the software checks

Code builds the geometry from tested components, then evaluates it against the desk, bottle, screws and other reference bodies.

Check	Meaning in the bottle-holder example
Fit and contact	The clamp leaves room for the desk and touches the required faces.
Insertion and retention	The bottle and screw can enter. Geometry blocks unwanted escape directions.
Strength and stiffness	A finite-element simulation estimates stress and bending under the chosen load.
Manufacturability	Checks cover minimum walls, printer bounds and resin drainage. PreForm checks supports.

Failed checks trigger repair. Code keeps the best checkpoint and rolls back regressions. Unverified blocking objections prevent approval.

Benchmark protocol and full results

Task: size a support beam in a simulated rig. Two fixture scenarios and two seeds give four runs per method, with the same allowed budgets.

Method	Passed	New design passed	Mean volume vs reference	AI tokens per run
Numerical optimizer	4/4	4/4	0.76	0
Collective	4/4	1/4	0.93	161k
Single agent, same tools	4/4	2/4	0.88	31k
Agents without evidence exchange	4/4	2/4	0.87	52k

Lower volume is better. All AI methods used Nemotron 3 Ultra. The collective consumed the most tokens and did not outperform the baselines.

This small simulated benchmark does not test open-ended part design or establish a general ranking.

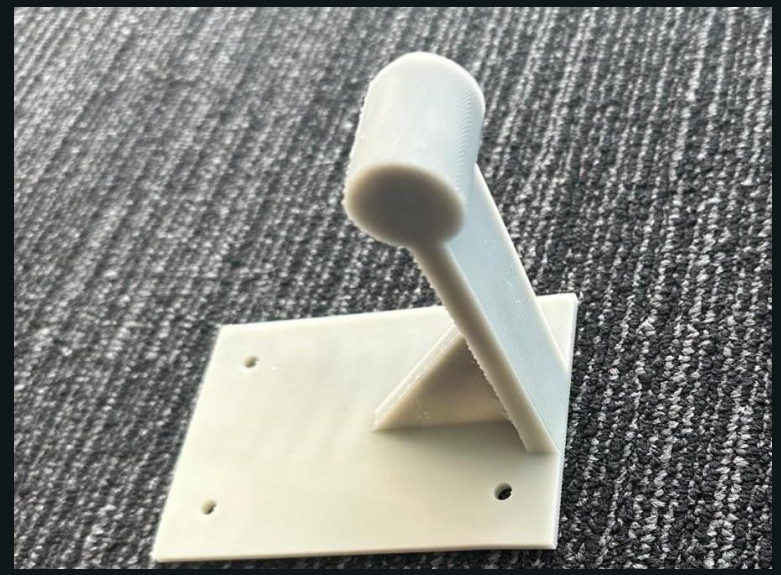
Print preparation and physical validation

PreForm orients and supports the part. Queueing requires every hard gate to pass, at least two agent approvals, and a user action.

PreForm result	Bottle holder	Backpack holder
Grade / support warning	WARN / 0.5% unsupported	WARN / 3.9% unsupported
Estimated print / resin	1.8 h / 157 mL	1.5 h / 117 mL
Agent approvals	3 of 3	3 of 3

Physical status: backpack print complete. No measured mass or bend yet.

The test plan is to match the modeled fixture and load, measure bend at the load point, then enter it in CoForm. Code updates effective stiffness and reruns the checks.



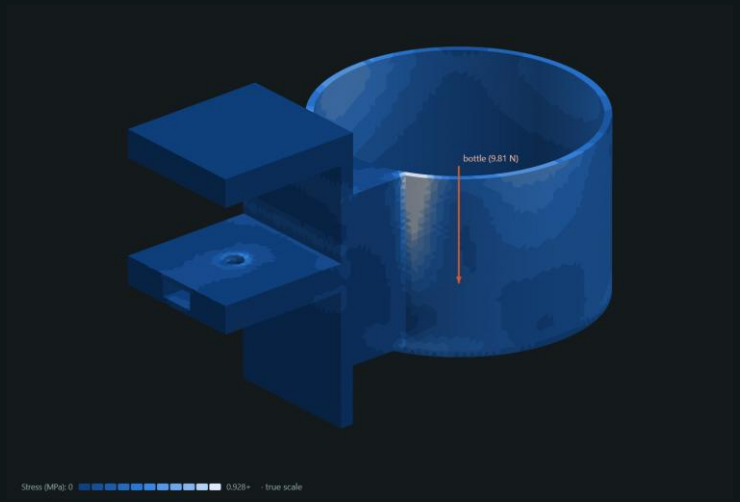
The printed backpack holder, before load testing

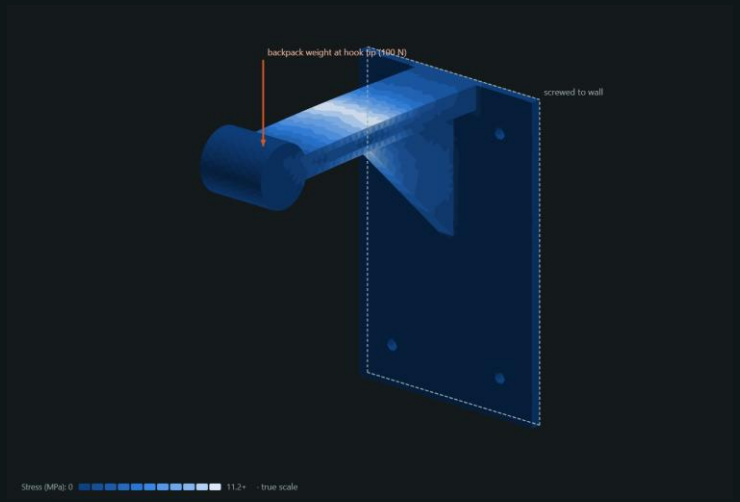
Evidence for the ScienceClaw judging rubric

Criterion	Evidence in this submission	Slides
Problem significance / 20%	A physical part must satisfy fit, strength and fabrication constraints.	1-2, 8
Technical impact / 25%	Generated parts with 16-48% modeled mass savings. One completed print.	5, 12
Decentralized agency / 20%	Separate role instructions, proposals and reviews. Code enforces shared rules.	3-4
Collective capability / 25%	Concrete design tradeoff plus single-agent and numerical baselines.	4, 6, 9
Execution and validation / 10%	Working pipeline, saved evidence, simulations and PreForm. Load test pending.	6, 8, 10

Official brief and rubric: scienceclawhack.ai

Other demonstrated designs and current scope





Result	Bottle holder	Earlier backpack holder
Predicted part mass	158.1 g to 125.4 g (-21%)	118.5 g to 99.1 g (-16%)
Predicted bend	0.08 mm under 1 kg	1.81 mm under 100 N
Fabrication status	Approved for printer queue	Print complete, no load test yet

Current catalogue: desk or wall mounts with cups, cradles or hooks. A recent portable phone-stand request has no accepted design in the saved record.